

ANDA, China

WMO: 508540

Lat: 46.3855N

Lon: 125.3240E

Elev: 150

StdP: 99.54

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-26.2	-32.0	0.2	-27.5	-30.0	0.2	-25.2	7.4	-13.1	6.6	-12.7	1.3	220	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	32.0	20.5	30.3	20.6	28.9	20.4	24.3	27.9	23.4	27.1	22.5	26.3	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.4	26.4	22.3	17.2	25.6	21.3	16.2	24.9	74.8	28.0	71.0	27.0	67.4	26.2	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.2	6.2	DB	-29.8	34.9	2.9	1.8	-31.9	36.2	-33.7	37.2	-35.3	38.2	-37.4	39.6	(4)
(5)			WB	-29.9	25.5	2.9	1.2	-32.0	26.3	-33.7	27.0	-35.4	27.7	-37.5	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-17.7	-12.6	-3.3	7.1	15.0	21.3	23.6	21.6	15.4	6.1	-5.8	-15.6	(6)	
	DBStd	15.21	4.39	5.37	5.90	4.91	4.19	3.45	2.39	2.55	3.78	4.76	6.09	4.67	(7)	
	HDD10.0	3432	858	632	412	114	6	0	0	0	4	137	474	794	(8)	
	HDD18.3	5383	1117	865	670	338	121	12	1	4	102	378	724	1053	(9)	
	CDD10.0	1491	0	0	1	27	161	338	421	359	167	18	0	0	(10)	
	CDD18.3	400	0	0	0	1	18	99	163	105	14	0	0	0	(11)	
	CDH23.3	3257	0	0	0	14	211	908	1291	712	119	3	0	0	(12)	
	CDH26.7	952	0	0	0	2	59	321	397	156	17	0	0	0	(13)	
(14) Wind	WSAvg	2.8	2.1	2.5	3.2	3.7	3.6	2.9	2.6	2.4	2.6	2.7	2.6	2.2	(14)	
(15) Precipitation	PrecAvg	429	1	2	5	15	30	66	142	97	50	14	4	2	(15)	
	PrecMax	680	5	10	25	98	107	144	284	224	114	42	15	10	(16)	
	PrecMin	292	0	0	0	1	2	14	49	22	9	0	0	0	(17)	
	PrecStd	82	1	2	5	19	23	32	57	58	25	12	4	2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.6	4.8	15.4	25.2	31.3	35.0	34.3	31.5	28.6	22.4	11.5	0.8	(19)	
		MCWB	-6.0	-0.2	6.8	12.3	16.1	19.3	21.9	22.5	19.2	13.7	6.5	-0.9	(20)	
	2%	DB	-6.9	1.4	11.4	21.7	28.4	32.2	32.0	29.8	26.3	19.0	8.6	-4.0	(21)	
		MCWB	-8.7	-2.1	4.0	10.3	15.0	19.2	21.7	22.3	18.2	12.1	4.1	-5.7	(22)	
	5%	DB	-9.3	-1.2	8.6	18.9	25.8	30.2	30.4	28.4	24.5	16.2	5.4	-6.4	(23)	
		MCWB	-10.7	-4.1	2.3	9.1	14.5	18.8	22.1	21.5	16.8	10.0	1.9	-7.9	(24)	
	10%	DB	-11.1	-3.9	5.8	16.1	23.2	28.3	28.8	27.1	22.7	14.0	2.5	-8.7	(25)	
		MCWB	-12.2	-6.3	0.5	7.7	13.1	18.4	22.0	20.8	15.8	8.7	-0.3	-9.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.9	0.2	7.1	13.2	18.2	22.7	26.2	25.2	21.7	15.2	7.6	-1.2	(27)	
		MCDB	-3.9	3.6	14.4	22.8	26.5	28.9	29.4	28.9	25.6	19.7	10.9	0.5	(28)	
	2%	WB	-8.6	-1.9	4.6	11.3	16.7	21.6	24.8	24.1	19.7	12.9	4.4	-5.6	(29)	
		MCDB	-6.9	1.4	10.2	19.7	24.4	27.8	28.3	27.7	23.9	17.8	7.9	-4.2	(30)	
	5%	WB	-10.6	-4.2	2.7	9.7	15.5	20.7	23.9	23.0	18.3	11.2	2.1	-7.9	(31)	
		MCDB	-9.2	-1.3	8.1	17.9	22.8	26.4	27.6	26.3	22.7	15.2	5.2	-6.5	(32)	
	10%	WB	-12.2	-6.3	0.8	8.1	14.2	19.8	23.1	22.1	17.0	9.2	-0.1	-9.9	(33)	
		MCDB	-11.1	-4.0	5.5	15.3	22.0	25.4	26.8	25.5	21.3	13.2	2.2	-8.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.1	10.8	11.5	11.1	9.9	8.4	9.1	11.1	10.3	8.9	9.1	(35)	
		MCDBR	11.3	12.4	14.0	15.4	14.7	13.0	10.9	10.8	13.2	13.6	11.6	10.6	(36)	
	5% WB	MCWBR	10.0	9.6	8.5	8.1	6.2	4.5	4.1	4.6	6.1	7.7	8.1	9.1	(37)	
		MCDWR	11.2	12.2	13.1	13.6	11.6	9.6	8.1	8.3	10.6	12.0	11.2	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.308	0.326	0.398	0.495	0.559	0.608	0.565	0.501	0.446	0.422	0.351	0.319	(40)	
		taud	2.155	2.135	1.997	1.822	1.740	1.731	1.872	2.015	2.065	2.010	2.124	2.140	(41)	
	Ebn at Noon		758	827	809	760	726	693	718	750	754	695	689	689	(42)	
		Edh at Noon	89	112	150	197	221	225	193	161	140	127	92	81	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.05	3.21	4.38	5.20	5.81	5.82	5.47	5.13	4.43	3.09	2.01	1.58	(44)	
	RadStd		0.23	0.21	0.30	0.35	0.45	0.54	0.54	0.35	0.36	0.17	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page